

TrES-3b

R-1241

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_200528 · created 2026-07-31T11:01:02+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2458997.903 +/- 0.0012 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.196 +/- 0.015
Transit depth (Rp/Rs)^2	3.85 +/- 0.59 [%]
Orbital Inclination (inc)	81.85 +/- 0.39
Airmass coefficient 1 (a1)	0.9356 +/- 0.008
Airmass coefficient 2 (a2)	0.062 +/- 0.029
Scatter in the residuals of the lightcurve fit is	0.86 %
Best Comparison Star	#3 - [348, 321]
Optimal Aperture	3.95
Optimal Annulus	6.65
Transit Duration (day)	0.0585 +/- 0.0038

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.196 ± 0.015 vs 0.16309 (deviation 1.73σ)
Duration fit vs published	0.0585 d vs 0.0567 d (deviation 0.38σ)
Mid-time O–C	1.5 min
Depth SNR	10.3
Residual scatter	0.86 %

Light curve

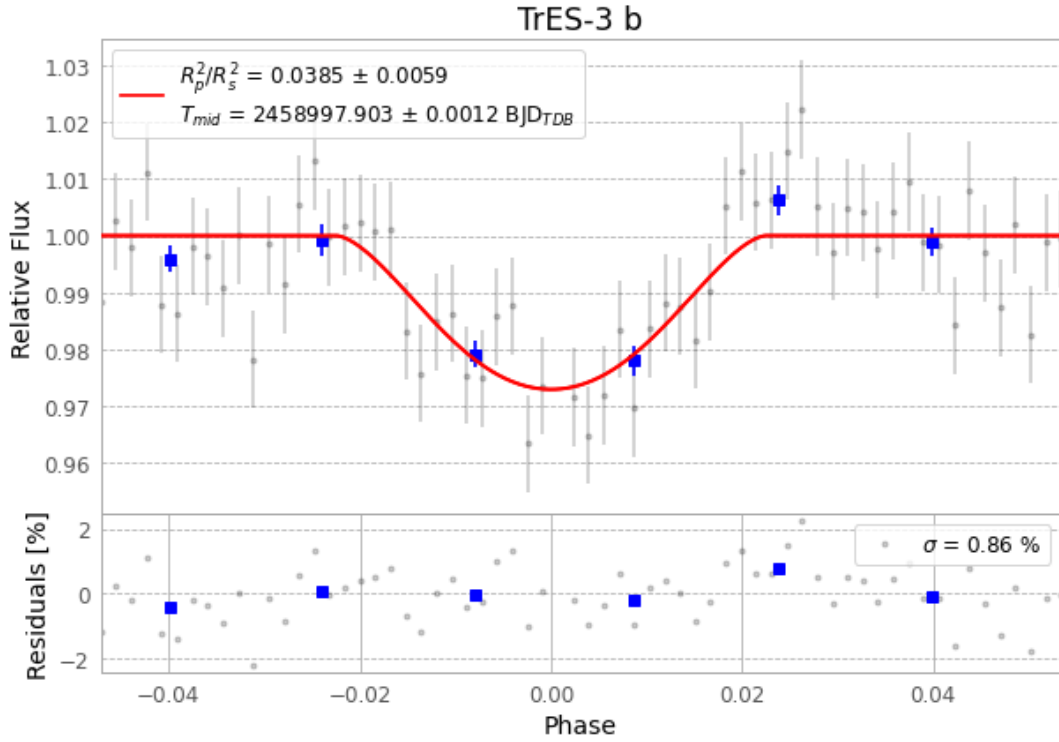


Figure 1 — FinalLightCurve_TrES-3 b_2020-05-28.png (SHA-256 4947bb9fe59953e7...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [419, 237], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 00207fdc1394fa8f...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

64 FITS frames (42 MB), TRES-3200528080612.FITS → TRES-3200528111512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-200528.log (6ed114bd2fac...), FinalLightCurve_TrES-3 b_2020-05-28.png (4947bb9fe599...), FinalLightCurve_TrES-3 b_2020-05-28.csv (56d09e08563a...)

Compute resources

Wall time 210.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-31 11:01Z.